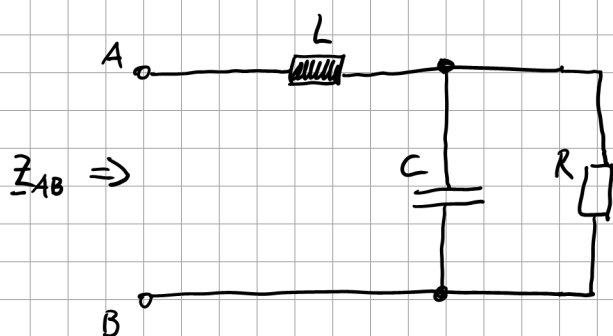


YZ 4



$$Z_{AB} = j\omega L + \frac{\frac{1}{j\omega C} \cdot R}{\frac{1}{j\omega C} + R} = j\omega L + \frac{\frac{R}{j\omega C}}{1 + \frac{j\omega RC}{j\omega C}}$$

$$= j\omega L + \frac{R}{1 + j\omega RC} = \frac{j\omega L (1 + j\omega RC) + R}{1 + j\omega RC}$$

$$= \frac{j\omega L - \omega^2 RCL + R}{1 + j\omega RC}$$

$$\frac{1 - j\omega RC}{1 - j\omega RC}$$

$$\underline{z} \cdot \underline{z}^* = |\underline{z}|^2$$

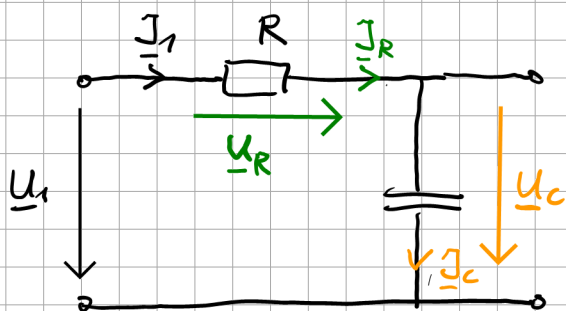
$$= \frac{j\omega L - \omega^2 RCL + R}{1 + (\omega RC)^2} + \frac{\omega^2 RLC + j\omega^3 R^2 C^2 L - j\omega R^2 C}{1 + (\omega RC)^2}$$

$$= \frac{R}{1 + (\omega RC)^2} + j \frac{\omega L - \omega R^2 C + \omega^3 R^2 C^2 L}{1 + (\omega RC)^2}$$

5 Ströme und Spannungen in linearen Netzwerken bei Sinusanregung

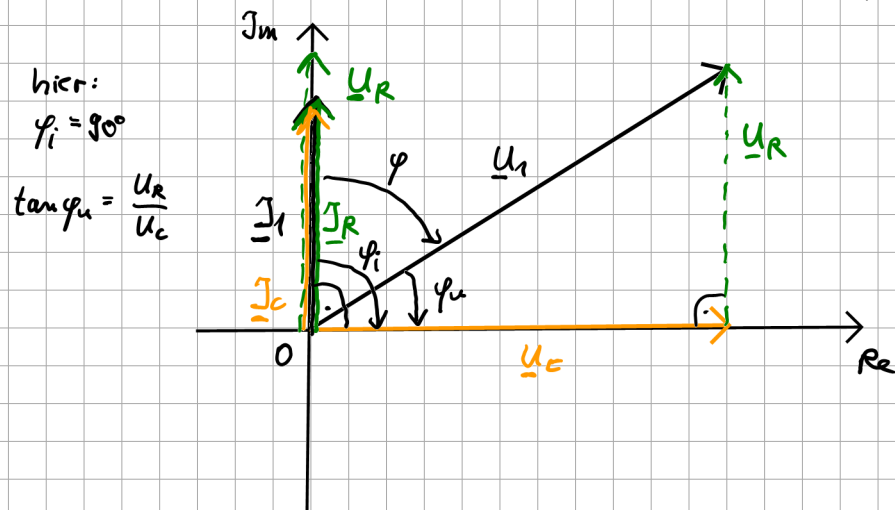
„linear“ : R, L, C

5.1 R-C-Schaltung



$$\underline{I}_1 = \underline{I}_R = \underline{I}_C$$

$$\varphi = \varphi_u - \varphi_i$$



$$\underline{U}_R = R \cdot \underline{I}_R$$

$$\underline{U}_C = \frac{1}{j\omega C} \underline{I}_C$$

$$= \frac{1}{j} \cdot \frac{1}{\omega C} \underline{I}_C$$

$$\frac{1}{j} = e^{-j90^\circ}$$

Pythagoras : $|\underline{U}_1|^2 = |\underline{U}_C|^2 + |\underline{U}_R|^2$

$$U_1^2 = U_C^2 + U_R^2$$

$$U_1 = \sqrt{U_C^2 + U_R^2}$$

$$\left| \frac{a}{b} \right| = \frac{|a|}{|b|}$$

$$|a \cdot b| = |a| \cdot |b|$$

$$\underline{U}_C = \frac{1}{j\omega C} \cdot \underline{I}_1$$

$$U_C = |\underline{U}_C| = \left| \frac{\underline{I}_1}{j\omega C} \right| = \frac{|\underline{I}_1|}{|j\omega C|} = \frac{I_1}{|j| \cdot |\omega C|} = \frac{I_1}{\omega C}$$

$$\underline{U}_R = R \cdot \underline{I}_1$$

$$U_R = R \cdot I_1$$

$$U_1 = \sqrt{\left(\frac{I_1}{\omega C} \right)^2 + (R \cdot I_1)^2}$$

$$U_1 = I_1 \cdot \sqrt{\underbrace{\left(\frac{1}{\omega C}\right)^2}_{X_C} + R^2} = I_1 \cdot \sqrt{X_C^2 + R^2}$$

$$I_1 = \frac{U_1}{\sqrt{X_C^2 + R^2}}$$

$$\varphi = \varphi_u - \varphi_i$$

$$\tan \varphi_u = \frac{U_R}{U_C} \Rightarrow \varphi_u = \arctan \frac{U_R}{U_C}$$

$$\varphi_i = 90^\circ$$

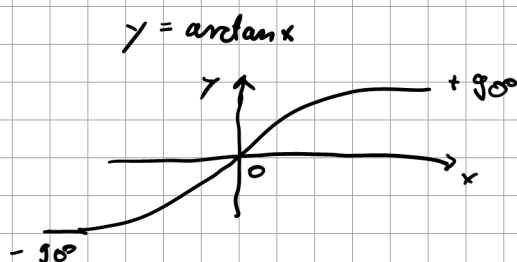
$$\varphi = \arctan \left(\frac{U_R}{U_C} \right) - 90^\circ$$

$$U_R = R \cdot I_1$$

$$U_C = \frac{1}{\omega C} \cdot I_1$$

$$\varphi = \arctan \left(\frac{R \cdot \cancel{I_1}}{\frac{1}{\omega C} \cdot \cancel{I_1}} \right) - 90^\circ$$

$$\varphi = \arctan(\omega RC) - 90^\circ$$



Lösungsweg: Komplexe Rechnung

$$\underline{U}_1 = \left(R + \frac{1}{j\omega C} \right) \underline{I}_1$$

$$\Rightarrow \underline{I}_1 = \frac{\underline{U}_1}{R + \frac{1}{j\omega C}} = \frac{\underline{U}_1}{\frac{j\omega CR + 1}{j\omega C}}$$

$$\underline{I}_1 = \frac{j\omega C}{1 + j\omega RC} \cdot \underline{U}_1$$

$$|\underline{I}_1| = I_1 = \left| \frac{j\omega C}{1 + j\omega RC} \cdot \underline{U}_1 \right| = \frac{|j\omega C|}{|1 + j\omega RC|} \cdot |\underline{U}_1|$$

$$\underline{I}_1 = \frac{\omega C}{\sqrt{1 + (\omega RC)^2}} \cdot U_1$$

$$\omega C = \frac{1}{\frac{1}{\omega C}} =$$

$$= \frac{1}{\sqrt{\frac{1}{(\omega C)^2} + \frac{(\omega RC)^2}{(\omega C)^2}}} \cdot U_1$$

$$= \frac{1}{\sqrt{\underbrace{\left(\frac{1}{\omega C}\right)^2}_{X_C} + R^2}} U_1$$

$$\underline{I}_1 = \frac{j\omega C}{1 + j\omega RC} \underline{U}_1 \quad \Rightarrow \quad \frac{\underline{U}_1}{\underline{I}_1} = \frac{1 + j\omega RC}{j\omega C}$$

Wir suchen: $\varphi = \varphi_u - \varphi_i$

$$\angle \left(\frac{\underline{U}_1}{\underline{I}_1} \right) = \angle \left(\frac{1 + j\omega RC}{j\omega C} \right)$$

$$\underline{z} = \frac{a + jb}{c + jd} = \frac{\sqrt{a^2 + b^2} \cdot e^{j \arctan \frac{b}{a}}}{\sqrt{c^2 + d^2} \cdot e^{j \arctan \frac{d}{c}}}$$

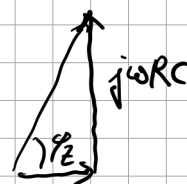
$$= \text{"} \cdot e^{j \underbrace{\arctan \left(\frac{b}{a} \right)}_{\varphi_{\text{Zähler}}} - \underbrace{\arctan \left(\frac{d}{c} \right)}_{\varphi_{\text{Nenner}}}}$$

$$\Rightarrow \angle(\underline{z}) = \text{Winkel Zähler} - \text{Winkel Nenner} \\ = \varphi_z - \varphi_n$$

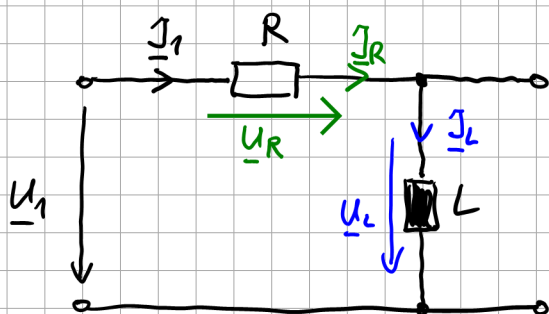
$$\angle \left(\frac{\underline{U}_1}{\underline{I}_1} \right) = \angle \left(\frac{1 + j\omega RC}{j\omega C} \right) = \varphi_z - \varphi_n$$

$$= \arctan \left(\frac{\omega RC}{1} \right) - 90^\circ$$

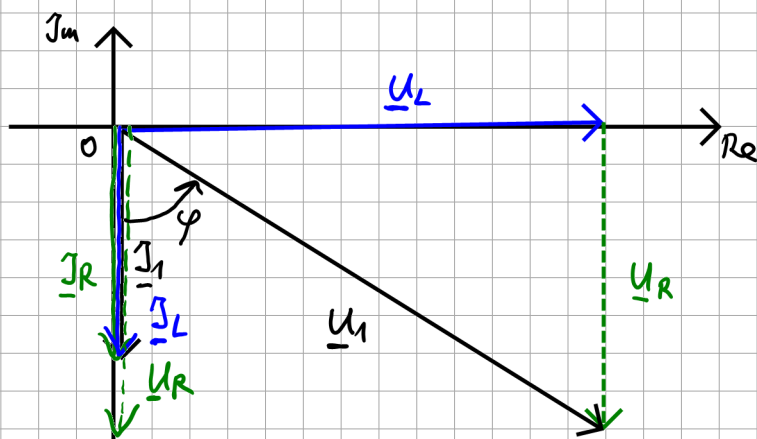
$$= \arctan(\omega RC) - 90^\circ < 0$$



5.2 R-L-Schaltung



$$\underline{I}_1 = \underline{I}_R = \underline{I}_L$$



$$\underline{U}_R = R \cdot \underline{I}_R$$

$$\underline{U}_L = j\omega L \cdot \underline{I}_L$$

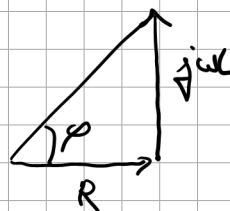
$$\underline{U}_1 = (R + j\omega L) \underline{I}_1$$

$$\underline{I}_1 = \frac{\underline{U}_1}{R + j\omega L}$$

$$I_1 = |\underline{I}_1| = \left| \frac{\underline{U}_1}{R + j\omega L} \right| = \frac{U_1}{\sqrt{R^2 + (\omega L)^2}} = \frac{1}{\sqrt{R^2 + \underbrace{(\omega L)^2}_{X_L^2}}} \cdot U_1$$

$$I_1 = \frac{1}{\sqrt{R^2 + X_L^2}} U_1$$

nach
von $\frac{\underline{U}_1}{\underline{I}_1} = R + j\omega L$



$$\angle \left(\frac{\underline{U}_1}{\underline{I}_1} \right) = \angle (R + j\omega L) = \arctan \left(\frac{\omega L}{R} \right) > 0$$